

Education

Degree	Specialization	Institute	Year	CPI/%
B.Tech-M.Tech (Dual-Degree)	Mechanical Engineering	Indian Institute of Technology Gandhinagar (IITGN)	2022–Present	8.48/10
Class XII	Physics, Chemistry, Maths, Computer Science	Delhi Public School, Siliguri	2020–2021	90.00%
Class X	—	Delhi Public School, Siliguri	2018–2019	95.40%

Research Affiliations & Appointments

Position	Research Focus	Institution	Period
Visiting Researcher	Foreign <i>Bimanual Manipulation, Robot Control, Contact-Rich Manipulation, Robot Learning</i>	Smart Robots Design Lab, Tohoku University, Japan	May 2026–Mar 2027
Student Researcher	<i>Bimanual Manipulation, Redun- dancy Optimization, Robot Learning and Control</i>	IITGN Robotics Lab, IIT Gandhina- gar, India	Dec 2023–Present
Visiting Researcher	<i>Bimanual Manipulation, Dynam- ical Systems Control, Humanoid Robotics</i>	Advanced Robotics Team, Addverb Technologies, India	Dec 2024–Present

Publications and Preprints

- [1] **D. Das**, S. Jain, R. Kumar, and H. J. Palanhandalam-Madapusi, “Consensus-Driven Dynamical Systems Control for Dual-Arm Handover”, Proceedings of the **IEEE International Conference on Robotics and Automation (ICRA)**, 2026, Vienna, Austria. Earlier version presented as a Late Breaking Report (LBR) at the **IEEE-RAS International Conference on Humanoid Robots (Humanoids)**, 2025, Seoul, Korea. [[Supplementary Video](#)], [[Project Website](#)], [[Humanoids LBR Poster](#)]
- [2] **D. Das**, S. Barat, and H. J. Palanhandalam-Madapusi, “Task-Specific Manipulability Metrics for Redundancy Optimization in Cooperative Manipulation”, **accepted for publication in Industrial Robot: The International Journal of Robotics Research and Application**, 2026. DOI: [10.1108/IR-05-2026-0221](https://doi.org/10.1108/IR-05-2026-0221). [[Project Website](#)], [[Supplementary Video](#)]
- [3] **D. Das**, A. A. Ravankar, Y. Hirata, and H. J. Palanhandalam-Madapusi, “TwistGrasp: Predicting Feasibility and Torque Capability for Continuous Unscrewing”, submitted to the **IEEE International Conference on Robotics and Automation (ICRA)**, 2027. [[Supplementary Video](#)]
- [4] S. Kumar, S. Barat, **D. Das**, S. Jain, R. Kumar, and H. J. Palanhandalam-Madapusi, “Towards Coordinated Dual-Arm Snap-Fit Assembly Skill for Delicate Applications”, **under review at IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2025. Preprint available at [arXiv:2511.18153](https://arxiv.org/abs/2511.18153). Earlier version presented as a Late Breaking Report (LBR) at the **IEEE-RAS International Conference on Humanoid Robots (Humanoids)**, 2025. [[Project Website](#)], [[Poster](#)]
- [5] S. Barat, V. K. Joshi, **D. Das**, and H. J. Palanhandalam-Madapusi, “Grasp Dexterity Index: Manipulability of Objects Grasped by Multi-fingered Robotic Hands”, manuscript submitted to the **International Journal of Humanoid Robotics**, 2026. Preprint available at [doi:10.21203/rs.3.rs-9620791/v1](https://doi.org/10.21203/rs.3.rs-9620791/v1).
- [6] S. Barat, S. Patidar, **D. Das**, S. Kumar, S. Jadav, and H. J. Palanhandalam-Madapusi, “Geometric Transfer and Few-Trial Adaptation of Iteratively Learned Control Inputs for Robotic Manipulation Against Unknown Resistance Fields”, **under review at Industrial Robot: The International Journal of Robotics Research and Application**, 2026. [[Supplementary Video](#)]
- [7] **D. Das**, H. Ruparel, D. Gadhave, and H. J. Palanhandalam-Madapusi, “Dynamical-System Coordination Primitives for Bimanual Manipulation via Semantic Grounding”, Manuscript in preparation for submission to **IEEE Transactions on Robotics (T-RO)**, 2026.

Academic Research

Smart Robots Design Lab, Tohoku University

Sendai, Miyagi, Japan

Robot Control and Learning for Contact-Rich Bimanual Manipulation

May 2026 – Mar 2027

Visiting Foreign Researcher

JST LOTUS Programme

Host Advisor: Prof. Ankit A. Ravankar, Department of Robotics, Graduate School of Engineering, Tohoku University

- Selected as a **Visiting Foreign Researcher** under the **JST LOTUS / India–Japan Circulation of Talented Youths in Science Programme** for a long-term research visit at Tohoku University.
- Developed **TwistGrasp**, a model-based framework for **task-conditioned grasp selection in contact-rich bimanual manipulation**, introducing **ScrewHorizon** and **ScrewCapability** to predict remaining motion feasibility and future task-axis torque capability under joint-limit, collision, and task-directed rate constraints.
- Implemented and validated the framework in **MuJoCo and on a physical dual-UR5e system**, including screw-space motion generation, grasp evaluation, autonomous reconfiguration, and bottle unscrewing, forming the **first-author work listed in [3]**.
- Currently investigating how **tactile and force feedback** can be integrated with structured robot controllers for contact-rich manipulation, including **residual reinforcement learning** to adapt model-based policies to contact uncertainty and unmodeled interaction dynamics.
- Extending this work toward a broader family of **contact-rich bimanual manipulation skills**, including container opening, pouring, precision handling, and actor–stabilizer tasks, with emphasis on reusable control primitives and learning from physical interaction.

IITGN Robotics Lab [🔗](#), IIT Gandhinagar

Gandhinagar, Gujarat, India

Task-Specific Manipulability and Redundancy Optimization for Cooperative Manipulation

July 2024 – Oct 2024

[Project Link](#) [🔗](#)

Undergraduate Researcher

Advisor: Prof. Harish Palantheandalam–Madapusi [🔗](#), IITGN Robotics Lab

- Developed a **closed-loop inverse-kinematics and null-space redundancy-resolution framework** for cooperative manipulation, integrating grasp compatibility and hand-Jacobian kinematics to shape redundant robot configurations without disturbing the commanded object motion.
- Formulated and compared **velocity, force, and directional-force manipulability objectives**, and validated them on a **planar dual-arm hardware platform** under external loading using actuator-current-based effort measurements.
- Extended the framework in **MuJoCo with two 7-DoF Franka Panda manipulators** across static holding, translational pick-and-place, and **six-dimensional pick-and-place** from multiple initial configurations, resulting in the **accepted journal publication [2]**.

Task-Based Generalization for Iterative Learning Controllers (ILCs)

Dec 2023 – June 2024

Research Intern, Winter '23 | Summer Research Internship Programme (SRIP) '24

[Project Link](#) [🔗](#)

Advisor: Dr. Harish Palantheandalam–Madapusi [🔗](#), Professor, Mechanical Engineering, IITGN

- Developed an **Iterative Learning Control (ILC)** framework that enables robots to **generalize manipulation behaviors across arbitrary trajectories**, integrating **virtual system dynamics** and **feedback linearization** within an admittance-based control scheme for adaptive interaction.
- Experimentally validated on hardware through (i) **adaptive cutting of unknown materials** and (ii) **bimanual planar grasping of objects with unknown mass-inertia**.
- Implemented an **Extended Kalman Filter (EKF)** to fuse high-rate proprioceptive and low-rate visual feedback for accurate object state estimation.

Snap-Fit Assembly and Event-Driven Compliance for Multi-Arm Coordination

Feb 2025 – July 2025

Undergraduate Researcher

[Video](#) [🔗](#)

Advisor: Prof. Harish Palantheandalam–Madapusi [🔗](#), IITGN Robotics Lab, IIT Gandhinagar

- Contributed to a unified dual-arm snap-fit framework for lens–frame insertion, integrating, **compliant control**, and **multi-arm coordination**.
- Designed **SnapNet** (CNN–GRU–attention) to detect the instant of mechanical engagement (“snap”) from joint velocities in real time.
- Enabled **event-triggered impedance gain modulation** to introduce compliance during insertion, reducing over-insertion and part damage.

Industrial Research

Advanced Robotics Team, [Addverb Technologies Limited](#)

Noida, Uttar Pradesh, India

Consensus-Driven Dynamical Systems Control for Dual-Arm Handover

April 2025 – July 2025

Visiting Researcher

[Project Link](#)

Advisor: [Dr. Rajesh Kumar](#), Principal Scientist (Robotics & AI)

- Designed a **dynamical-systems bimanual handover controller** that enables **phase-locked, obstacle-aware, real-time** handovers by unifying asynchronous and synchronous motion via **coupled translational & quaternion DS laws**.
- Collaborated with the industrial R&D team to deliver **deployable solutions** for humanoid and bimanual platforms, culminating in the **publication in [1]**.

Coordinated Bimanual Transport of Large Objects

Dec 2024 – Feb 2025

Visiting Researcher

June 2025 – July 2025

Advisor: [Dr. Rajesh Kumar](#), Principal Scientist (Robotics & AI)

[Project Link](#)

- Integrated a **dynamical-systems-based control framework** for coordinated dual-arm transport, achieving **real-time synchronization** and **object-level control** for robust manipulation of large payloads.
- Validated the framework on a **humanoid robot** and developed deployable control modules for the industrial R&D team for their humanoid and bimanual systems.

Reasoning-Conditioned DS Primitives for Bimanual Manipulation

Sept 2025 – Present

Visiting Researcher

Advisor: [Dr. Rajesh Kumar](#), Principal Scientist (Robotics & AI)

- Developed a **modular bimanual manipulation framework** comprising **task-specific DS primitives** that can be triggered to execute diverse bimanual skills such as handover, transport, and assembly.
- Integrating an **LLM-VLM-based autonomy stack** to bridge classical DS control with learning-based reasoning, keeping high-level decision-making neural while preserving model-based stability at the control layer.

Projects

SciML for Chaotic Dynamics: Learning Double-Pendulum Trajectories (LNN & SINDy)

Sep 2025 – Nov 2025

Course Project – ME 691 XVII: Scientific Machine Learning for Thermo-Fluids

[Project Link](#)

Advisor: Prof. Shubhangi Bansode, IIT Gandhinagar

- Built a MuJoCo-based data pipeline for planar double-pendulum rollouts under **ideal (conservative)** and **stiction-dominated** regimes; performed windowing, downsampling, normalization, and derivative extraction for supervised learning.
- Trained **Lagrangian Neural Networks (LNNs)** with Euler-Lagrange structure, and extended them with a **learned dissipation term** to model frictional decay without hand-coded friction laws.
- Implemented a **physics-aware SINDy / PySINDy** library (trig + velocity-coupling features) and compared models via **long-horizon rollouts, energy behavior diagnostics, and inference-time trade-offs**.

Null-Space Manipulability Optimization (Franka, MuJoCo)

Oct 2025

Course Project – ES 604: Engineering Optimization

[Project Link](#)

Advisor: Prof. Hari Ganesh

- Formulated redundancy optimization using a **D-optimality objective** to improve dexterity while preserving end-effector pose tracking.
- Implemented a **Damped Least Squares (DLS)** velocity tracker with a **Quadratic Programming (QP) in joint velocities** (pose tracking + regularization + joint-limit box constraints) running in MuJoCo (Franka Panda Robot).
- Compared **velocity-space null-space optimizers** (Steepest Ascent + Armijo, PR+ Conjugate Gradient, L-BFGS with line-search), showing **consistent manipulability improvement** and **faster convergence** versus a no-null-space baseline.

Social Robotics: Proxemics, Interaction & Communication Algorithms for HRI

Sept 2024 – Oct 2024

Course Project – ME 692: Social Robotics

Advisor: Prof. Barbara Bruno, KIT Germany

- Implemented **proxemics estimators** using monocular & stereo vision to evaluate interpersonal distances in HRI.
- Developed **gesture recognition & attention estimation models** using MediaPipe for real-time interaction analysis.
- Integrated **verbal communication modules** for speech recognition & dialogue management using PyTorch & Gemma2B.

Visual Deception & Sensory Deprivation Effects on Human Arm Impedance

March 2024 – April 2024

Course Project – ES 656: Human-Robot Interaction

[Project Link](#)

Advisor: Prof. Vineet Vashista, Human Centered Robotics Lab, IIT Gandhinagar

- Investigated the influence of **visual and sensory cues on human arm impedance** for improved human-robot cooperation.
- Analyzed stiffness and damping variations using impedance ellipsoids under varying feedback conditions.
- Studied how **deceptive perturbations affect sensorimotor adaptability** and impedance modulation.

Bio-Mechanical Movement Analysis using OpenSim

Feb 2024 – March 2024

Course Project – ES 656: Human-Robot Interaction

[Project Link](#)

Advisor: Prof. Vineet Vashista, Human Centered Robotics Lab, IIT Gandhinagar

- Collected and pre-processed squat motion data using a **Vicon motion capture system**.
- Applied **OpenSim** for inverse kinematics, dynamics, and optimization to study muscle activation and joint torques.

Trajectory Control Framework for a 6-DoF Robot

Dec 2023

Research Project – IITGN Robotics Lab & Orangewood Automation Pvt. Ltd.

[Project Link](#)

Advisor: Prof. Harish Palanthenalam-Madapusi, IIT Gandhinagar

- Implemented a **trajectory control framework** using the **Time-Optimal Path Parameterization (TOPP-RA)** approach.
- Enabled precise joint-space and cartesian-space trajectory execution for a 6-DoF industrial manipulator.

Explorations with the 2R Manipulator: Simulation & Hardware

August 2023 – February

Course Projects – ME 639: Introduction to Robotics & ES 656: Human-Robot Interaction

2024

Advisor: Prof. Harish Palanthenalam-Madapusi & Prof. Vineet Vashista, IIT Gandhinagar

[Project Link](#)

- Conducted simulation and hardware studies on **motor-actuated and cable-driven 2R manipulators**.
- Performed **workspace analysis, cable redundancy assessment, and trajectory-tracking experiments**.

Wearable Child Abuse Detection & Monitoring System

May 2023 – June 2023

Team Lead – Summer Research Project

[Project Link](#)

Advisor: Prof. Udit Bhatia, IIT Gandhinagar

- Led a team of **30 students** to design a **wearable IoT device ecosystem** for real-time child abuse detection using FSR data.
- Integrated machine-learning models and a mobile app interface; project received **national media coverage**.

Technical Skills

- **Robot Control & Manipulation:** Dynamical Systems Control, Impedance/Admittance Control, Redundancy Resolution & Null-Space Control, Differential IK, DLS, Quadratic Programming, Manipulability & Task-Space Analysis.
- **Programming & Robotics Middleware:** Python, C++, C, MATLAB/Simulink, ROS 1, ROS 2.
- **Simulation & Robotics Frameworks:** MuJoCo, Isaac Sim, Isaac Gym, Gazebo, robomimic, MSC Adams.
- **Robotics & Machine Learning Libraries:** KDL, MoveIt, TOPP-RA, NumPy, SciPy, PyTorch, TensorFlow, Scikit-learn, Pandas, Matplotlib, MediaPipe.
- **Prototyping & Engineering Tools:** Microcontrollers, 3D Printing, Autodesk Fusion 360, Autodesk Inventor, Laser Cutting.

Invited Talks & Outreach

Workshop Instructor - [IEEE RAS Student Chapter, PDEU: MuJoCo Workshop](#)

Jan 2026

Topic: Hands-On Inverse Kinematics with MuJoCo

ROS Talk - [ROSCon India 2025](#)

Dec 2025

Talk: Why Robots Today Can Do Backflips But Still Can't Make a Vada Pav?

Alumni Speaker — [TEDxDPS Siliguri Youth](#)

March 2025

Topic: From Curiosity to Real Impact — Building a Future with Robotics, AI, Engineering, and Purpose

Open Source Contributions

Contributor — [Mink 0.0.11](#) (Inverse Kinematics Toolkit for MuJoCo)

Feb 2025

- Contributed to the public release of **Mink 0.0.11**, enhancing the Python toolkit for MuJoCo-based inverse kinematics and motion planning.
- Implemented **new single and dual-arm Franka Panda examples** demonstrating bimanual manipulation examples and motion planning pipelines.

Teaching Assistantship and Mentorship

Teaching Assistant – ME639: Introduction to Robotics

IIT Gandhinagar, School of Engineering

Instructor: Prof. Harish Palanhandalam-Madapusi (2024), Prof. Madhu Vadali (2025)

Aug – Nov 2024

Aug 2025 – Present

- Conducted weekly **ROS- and hardware-based lab sessions** using manipulators and quadrupeds, supporting both theory and implementation.
- Mentored and designed student projects and assignments in **simulation (MuJoCo, Gazebo)** and **real-robot control pipelines**, bridging kinematic modeling with practical deployment.

Summer Mentorship – Robotics Research Projects

IITGN Robotics Lab, IIT Gandhinagar

Mentor: Prof. Harish Palanhandalam-Madapusi

May 2025 – July 2025

- Mentored **four undergraduate research students** on end-to-end robotics pipelines covering MuJoCo simulation, ROS integration, teleoperation and data collection for imitation learning.

Fellowships, Awards & Grants

JST LOTUS Programme – Visiting Foreign Researcher Fellowship

Japan Science and Technology Agency (JST), Tohoku University

Funded long-term research stay at Tohoku University under the India–Japan Circulation of Talented Youths in Science Programme.

2026

JPY 2.64M (~US\$16.8K)

International Research & Conference Travel Grant

Indian Institute of Technology Gandhinagar

Travel support for research at **Tohoku University, Japan** (INR 50,000) and presentation at **IEEE ICRA 2026, Vienna** (INR 1,50,000).

2026

INR 2,00,000 (~US\$2.08K)

International Conference Travel Support

IIT Gandhinagar & IITGN Robotics Lab

Travel support for presenting research at **IEEE-RAS Humanoids 2025, Seoul, South Korea**, comprising INR 60,000 in institute support and INR 50,000 from the **IITGN Robotics Lab**.

2025

INR 1,10,000 (~US\$1.14K)

Extra-Curricular Activities

- **Model United Nations & Debating** April 2018 - Present
 - Presiding Officer at IIMUN; trained 1600+ students & led MUNs on global issues, honing public speaking & leadership.
- **Co-Founder, Alpha Geeks** (www.alphageeks.in) June 2019 - Present
 - Promoted AI, robotics, & coding through workshops; organized hands-on tech events for student innovators.
- **Sponsorship Core, Amalthea Tech Summit, IITGN** April 2023 - Dec 2023
 - Revamped summit sponsorship, achieving a 3x fundraising increase; initiated 'Women In Tech' panel with notable speakers.